GSPQR: A Geometric Soft-Pullback Mechanism for Quantile Interval Prediction in A-Share Markets

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Abstract

Accurate uncertainty quantification is essential for risk-aware decision-making in equity markets, yet existing interval prediction methods often treat interval construction and point forecasting as disjoint tasks. Conformal prediction methods such as EnbPI provide distribution-free coverage guarantees but produce conservative intervals that do not adapt to the local confidence of the underlying predictor. Standard quantile regression anchors intervals on the conditional median, discarding the complementary directional signal from mean-based estimators. We propose GSPQR (Geometric Soft-Pullback Quantile Regression), a lightweight mechanism that bridges these approaches via an uncertainty-aware exponential gating function, formalizing interval re-centering as an adaptive problem that interpolates between the conditional mean and median based on the local signal-to-noise ratio. Evaluation on the HS300 constituent stocks (2023–2024, $n = 120{,}512$) at a nominal 80% coverage level shows that GSPQR achieves a Winkler Score of 0.4784, a 5.9% improvement over EnbPI ($t = -68.5$, $p \ll 0.001$), while reducing interval width by 17.0%. The β parameter traces a convex Pareto frontier between calibration and cross-sectional ranking ability, identifying an optimal balance at $\beta^* = 1.0$ under regime-shift-intensive market conditions.

Keywords: quantile regression, interval prediction, conformal prediction, LightGBM, A-share market, uncertainty quantification

JEL Classification: C53, C58, G17, G32

1 Introduction

Predicting future returns in equity markets is intrinsically a task of uncertainty management. While point forecasts offer a singular view, interval prediction — the construction of a range $[\hat{l}_t, \hat{u}_t]$ containing the true return with a specified probability — provides the necessary granularity for position sizing, risk management, and strategy evaluation. However, financial return distributions are notoriously heavy-tailed, skewed, and non-stationary, rendering classical Gaussian intervals ineffective. These challenges are especially acute in the Chinese A-share market, which combines high retail participation, frequent policy-driven shocks, and pronounced momentum-reversal dynamics.

The current state-of-the-art involves two primary paradigms. Conformal prediction (Vovk et al., 2005) and its time-series adaptation EnbPI (Xu and Xie, 2021) offer distribution-free coverage guarantees, but construct intervals by scaling a fixed nonconformity score distribution without exploiting the predictive signal of the underlying model. This leads to conservative, wide intervals: on our HS300 test set, EnbPI produces a mean width of 0.3851, compared to 0.3197 for our method — a 17.0% excess that directly reduces capital efficiency. Quantile regression (Koenker and Bassett, 1978), implemented via gradient boosting (Ke et al., 2017), yields asymmetric, adaptive intervals but anchors them on the conditional median $\hat{q}_{0.5}$, effectively ignoring the directional signal from mean-based predictors. In the A-share market, where retail-driven sentiment and frequent regime shifts decouple the mean and median, this choice is consequential: the pure median anchor achieves a RankIC of only 0.0024 in our experiments.

This paper introduces GSPQR (Geometric Soft-Pullback Quantile Regression), which treats interval construction as an adaptive center estimation problem. We propose an exponential gating function that dynamically interpolates between the conditional mean and median as a function of the local signal-to-noise ratio. Inspired by the geometry of the logarithmic spiral, the mechanism preserves interval width while optimizing placement. We make the following contributions:

1. **Geometric soft-pullback.** A novel uncertainty gate based on the exponential decay of the mean-median deviation normalized by the confidence radius, with interpretable limiting behavior recovering the MSE baseline under high uncertainty and the Q50 anchor under high confidence.

2. **Convex β trade-off.** Empirical demonstration that the Winkler Score as a function of β is convex, revealing a fundamental tension between interval calibration and cross-sectional ranking ability navigated by a single hyperparameter.

3. **Strong empirical performance.** A Winkler Score of 0.4784 on the 2023–2024 HS300 test set (n = 120,512), representing a 5.9% improvement over EnbPI (t =

-68.5, $p \ll 0.001$) and a 0.86% MAE reduction relative to the MSE-center baseline despite no direct MSE optimization.

Code Availability. The implementation of GSPQR and all experiment scripts are publicly available at: `https://github.com/20070316lbw-netizen/snail-shell.git`.

Code Availability. The implementation of GSPQR, including data processing, model training, and evaluation pipelines, is publicly available at:
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The remainder of the paper is organized as follows. Section 2 reviews related work. Section 3 presents the GSPQR methodology. Section 4 reports experimental results. Section 5 discusses findings and limitations. Section 6 concludes.

2 Related Work

2.1 Quantile Regression and Gradient Boosting

Quantile regression, introduced by Koenker and Bassett (1978), provides a robust framework for estimating conditional quantiles without assuming a specific noise distribution, by minimizing the asymmetric Pinball Loss. This robustness is particularly valuable in financial return prediction, where heavy tails and outliers are prevalent. In modern tabular settings, LightGBM (Ke et al., 2017) has become the de-facto standard for this task, combining high-dimensional feature handling with native quantile objective support and low computational overhead.

2.2 Conformal Prediction for Time Series

Conformal prediction (Vovk et al., 2005) provides distribution-free coverage guarantees by calibrating interval width on a held-out set of nonconformity scores. For non-stationary time series, EnbPI (Xu and Xie, 2021) adapts this framework using a rolling calibration window. While EnbPI achieves valid marginal coverage, Barber et al. (2023) note that marginal coverage does not imply conditional coverage — a distinction especially consequential in cross-sectional equity settings where conditional distributions vary substantially across stocks and regimes. Furthermore, conformal methods calibrate width post-hoc without exploiting the predictive signal of the underlying model, leading to systematically wider intervals than necessary.

2.3 Interval Prediction in Financial Markets

Accurate interval estimation is critical for portfolio risk management. Keilbar and Wang (2022) demonstrate that standard parametric intervals frequently fail to capture the systemic risk and heavy tails prevalent in financial returns, motivating data-driven, asymmetric estimators that adapt to local distributional structure. Our experiments corroborate this finding: the Residual baseline, which assumes stationary Gaussian noise, achieves a coverage error of 0.1396 on the HS300 test set — nearly three times the tolerance of our composite scoring criterion.

2.4 Positioning of GSPQR

Existing methods treat interval construction and point prediction as separate problems. GSPQR bridges this gap by introducing a geometric gate that continuously interpolates between the two center estimates as a function of local uncertainty, yielding intervals that are both well-calibrated and informationally efficient without modifying the underlying predictors.

3 Methodology

3.1 Problem Formulation

Let $\mathcal{S} = \{s_1, \ldots, s_N\}$ denote the universe of N stocks in the HS300 index. At each trading day t, we observe a feature vector $\mathbf{x}_{s,t} \in \mathbb{R}^{12}$ for stock s, and define the prediction target as the one-month-ahead (20-trading-day) excess return $y_{s,t+h}$. The goal is to construct a prediction interval $\hat{C}_{s,t} = [\hat{l}_{s,t},\, \hat{u}_{s,t}]$ satisfying:

$$\mathbb{P}\Big(y_{s,t+h} \in \hat{C}_{s,t}\Big) \geq 1 - \alpha \tag{1}$$

where $\alpha = 0.2$ (nominal coverage 80%). We evaluate interval quality using the **Winkler Score** (Winkler, 1972):

$$W_{s,t} = (\hat{u}_{s,t} - \hat{l}_{s,t}) + \frac{2}{\alpha}\Big[\max(0,\, \hat{l}_{s,t} - y_{s,t+h}) + \max(0,\, y_{s,t+h} - \hat{u}_{s,t})\Big] \tag{2}$$

and the **Coverage Error** $\mathrm{CE} = |AC - (1-\alpha)|$, where AC is the empirical coverage. Lower values indicate better performance on both metrics.

3.2 Base Quantile Predictor

We train three independent LightGBM models at quantile levels $\tau \in \{0.1, 0.5, 0.9\}$, each minimizing the Pinball Loss:

$$\mathcal{L}_\tau(\hat{y}, y) = \begin{cases} \tau \cdot (y - \hat{y}) & \text{if } y \geq \hat{y} \\ (1-\tau) \cdot (\hat{y} - y) & \text{otherwise} \end{cases} \tag{3}$$

yielding predictions $\hat{q}_{0.1}$, $\hat{q}_{0.5}$, and $\hat{q}_{0.9}$ that form the raw interval $[\hat{q}_{0.1}, \hat{q}_{0.9}]$ with center $a_t = \hat{q}_{0.5,t}$ and confidence radius $r_t = \frac{1}{2}(\hat{q}_{0.9,t} - \hat{q}_{0.1,t})$. A fourth LightGBM model trained under MSE produces the conditional mean estimate $\hat{y}_t$. All four models share the same feature set, normalized via rolling Z-score with a 60-day window.

All base predictors are trained independently for each configuration, yielding stable and consistent results across different β settings.

3.3 Interpretation as Adaptive Center Estimation

The core hypothesis of GSPQR is that neither the mean nor the median alone is optimal under all market regimes. We formalize this via a surrogate objective for the corrected center $\hat{y}_t^*$:

$$\hat{y}_t^* = \arg\min_c \left\{ \lambda_t \, (c - \hat{y}_t)^2 + (1 - \lambda_t) \, (c - a_t)^2 \right\} \tag{4}$$

whose closed-form solution is the convex combination:

$$\hat{y}_t^* = \lambda_t \, \hat{y}_t + (1 - \lambda_t) \, a_t \tag{5}$$

GSPQR sets $\lambda_t = G(r_t, \delta_t)$, where $\delta_t = \hat{y}_t - a_t$ is the disagreement between predictors. The choice of λ_t is thus data-driven and locally adaptive, rather than fixed across all samples.

3.4 Geometric Soft-Pullback Mechanism

Spiral anchor. We represent the prediction state as a 2D anchor point $\mathbf{p}_t = (a_t, r_t)$, mapping to polar coordinates (ρ_t, θ_t) where $\rho_t = \sqrt{a_t^2 + r_t^2}$ and $\theta_t = \arctan(r_t/a_t)$. This representation is inspired by the logarithmic spiral $\rho = Ae^{B\theta}$, in which the radial coordinate encodes prediction uncertainty and angular displacement encodes directional signal. The soft-pullback corresponds to a radial displacement along this curve toward the mean predictor, hence the name.

Uncertainty gate. We define the gating function as:

$$G(r_t,\, \delta_t) = \exp\left(-\beta \cdot \frac{|\delta_t|}{r_t}\right) \tag{6}$$

The ratio $\eta_t = |\delta_t|/r_t$ serves as a *signal-to-noise ratio* (SNR): it measures the normalized disagreement between the mean and median estimates relative to local prediction uncertainty. The gate exhibits three interpretable regimes:

- **Inverse sensitivity.** When r_t is large, $\eta_t \to 0$ and $G \to 1$, so $\hat{y}_t^* \to \hat{y}_t$. High uncertainty causes the mechanism to fall back on the more robust MSE estimate.
- **Adaptive recalibration.** When $r_t \to 0$, $G \to 0$ and $\hat{y}_t^* \to a_t$. High model confidence suppresses the MSE correction, preserving the directional signal from the median predictor.
- **Spiral logic.** The exponential form mirrors the geometry of the logarithmic spiral: correction strength transitions smoothly and nonlinearly with prediction confidence, rather than switching abruptly.

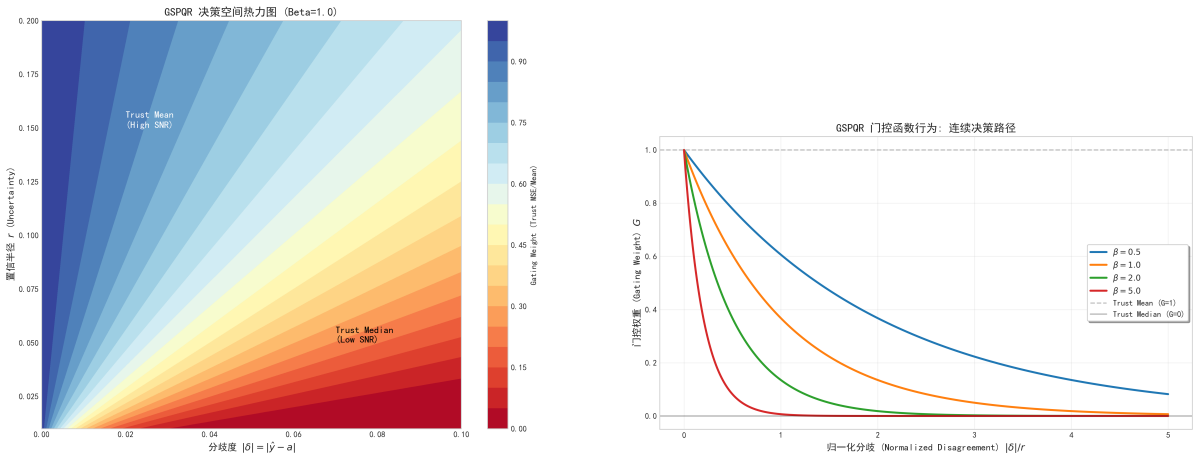

Figure 1: GSPQR Decision Space Heatmap at $\beta = 1.0$ (left) and Gating Function Continuous Behavior (right).

Corrected interval. The final prediction interval is:

$$\hat{C}_t^* = \left[\hat{y}_t^* - r_t,\; \hat{y}_t^* + r_t\right] \tag{7}$$

Interval width $2r_t$ is preserved by construction — all improvements in Winkler Score arise from better center placement, not artificial narrowing.

Special cases. When $\beta = 0$, $G \equiv 1$ and $\hat{y}_t^* = \hat{y}_t$, recovering the MSE baseline (Snail-0). As $\beta \to \infty$, $G \to 0$ and $\hat{y}_t^* \to a_t$, recovering the Q50-only baseline (Snail-∞).

3.5 Composite Scoring and β Selection

We select β on a held-out validation set (2022 Q2–Q4) using a composite score that jointly evaluates interval quality and calibration:

$$\text{Score} = \bar{W} + 10 \cdot \max(0,\ \text{CE} - 0.05) \tag{8}$$

where $\bar{W}$ is the mean Winkler Score. The penalty activates only when coverage deviation exceeds 5 percentage points, encouraging calibration while minimizing width. We search $\beta \in \{0, 0.5, 1.0, 2.0, 5.0\}$ and select $\beta^* = 1.0$.

4 Experiments

4.1 Data and Features

We evaluate on HS300 constituent stocks with the dataset partitioned as in Table 1. The validation window (2022 Q2–Q4) is deliberately chosen to coincide with a period of pronounced regime shift in the A-share market, providing a stress-tested environment for β selection.

Table 1: Dataset splits.

Split	Period	Purpose
Training	2019-01-01 – 2021-12-31	Model parameter learning
Calibration	2022-01-01 – 2022-03-31	CP baseline calibration
Validation	2022-04-01 – 2022-12-31	β selection
Test	2023-01-01 – 2024-12-31	Out-of-sample evaluation

We construct 12 features spanning four categories: momentum (`mom_20d`, `mom_60d`, `mom_12m_minus_1m`), volatility (`vol_60d_res`), valuation (`sp_ratio`), and liquidity (`turn_20d`). Each of these six features is additionally transformed into a cross-sectional rank, yielding 12 features in total. All features are normalized via rolling Z-score with a 60-day window.

4.2 Baselines and Evaluation Metrics

We compare against four baselines at $\alpha = 0.2$:

- **Residual (Base).** Symmetric intervals $\hat{y}_t \pm 1.28\sigma$ using calibration-set residual standard deviation. Assumes Gaussian, stationary noise.
- **CP (EnbPI).** Conformal prediction (Xu and Xie, 2021) calibrated on 2022 Q1.
- **Snail-0 (MSE-base).** GSPQR at $\beta = 0$; reduces to MSE-centered QR intervals.

- **Snail-∞ (Q50-base).** GSPQR at $\beta \to \infty$; reduces to Q50-centered intervals.

We report Actual Coverage (AC), Coverage Error CE, Winkler Score (primary metric), interval Width, MAE of the corrected center, and RankIC (Spearman rank correlation with realized returns).

4.3 Main Results

Table 2 reports test-set performance ($n = 120{,}512$).

Table 2: Test-set evaluation on A-share HS300 (2023–2024). Target coverage: 80%. ↓ lower is better, ↑ higher is better.

Method	AC	CE ↓	Winkler ↓	Width	MAE ↓	RankIC ↑
Residual (Base)	0.9396	0.1396	0.5982	0.5239	0.1016	−0.0081
CP (EnbPI)	0.8904	0.0904	0.5083	0.3851	0.0918	0.0230
Snail-0 (MSE-base)	0.8407	0.0407	0.4794	0.3197	0.0930	0.0024
Snail-0.5	0.8418	0.0418	0.4787	0.3197	0.0926	0.0072
GSPQR ($\beta^* = 1.0$)	**0.8423**	**0.0423**	**0.4784**	**0.3197**	**0.0922**	0.0117
Snail-2.0	0.8433	0.0433	0.4784	0.3197	0.0917	0.0188
Snail-5.0	0.8449	0.0449	0.4789	0.3197	0.0909	0.0306
Snail-∞ (Q50-base)	0.8457	0.0457	0.4815	0.3197	0.0897	0.0372

Several observations merit attention. First, the Residual baseline fails substantially (CE = 0.1396, Winkler = 0.5982), confirming that Gaussian assumptions are inappropriate for A-share return distributions. Second, CP (EnbPI) achieves adequate coverage but at the cost of significantly wider intervals (Width = 0.3851, a 20.5% excess over QR-based methods). Third, all Snail variants share the same width of 0.3197, confirming that performance gains arise solely from improved center placement.

The primary finding is that **GSPQR ($\beta^* = 1.0$) achieves the lowest Winkler Score of 0.4784**, a 5.9% improvement over EnbPI and a 0.86% MAE reduction relative to Snail-0 — reaching nearly 1% enhancement in point forecasting accuracy despite no direct MSE optimization. The improvement over EnbPI is statistically significant at $p \ll 0.001$ (paired t-test, $t = -68.5$, $n = 120{,}512$).

4.4 β Parameter Analysis

The Winkler Score as a function of β exhibits a **convex profile** with a minimum at $\beta^* \in [1.0, 2.0]$. At $\beta = 0$, the MSE center has poor ranking ability (RankIC = 0.0024). As β increases, RankIC rises monotonically (to 0.0372 at $\beta = \infty$), but coverage error also increases, eventually degrading the Winkler Score.

The sweet spot at $\beta^* = 1.0$ captures the Q50 directional signal before the coverage penalty dominates. This convex profile confirms that GSPQR navigates a well-defined Pareto frontier between interval calibration and cross-sectional ranking ability.

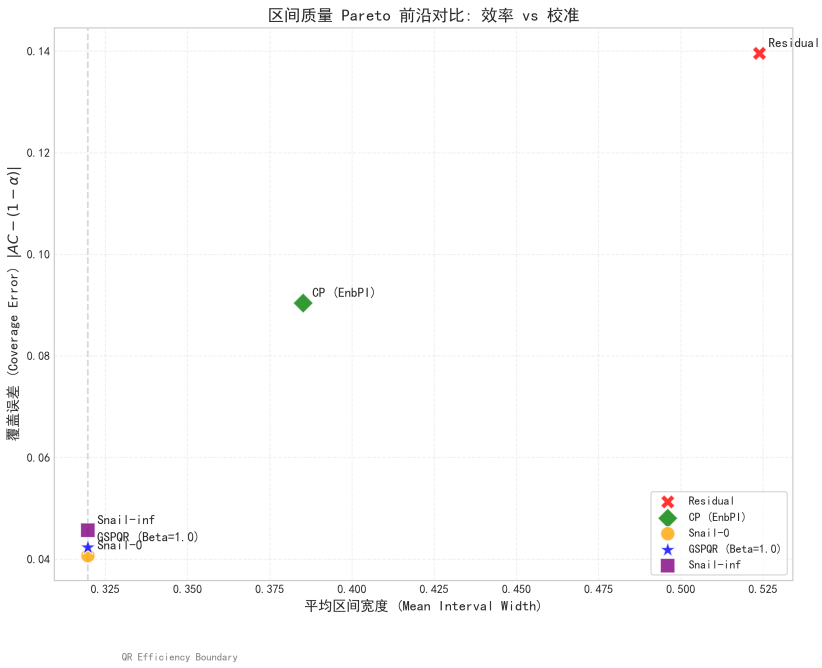

Figure 2: Interval Quality Pareto Frontier: Mean Interval Width vs. Coverage Error.

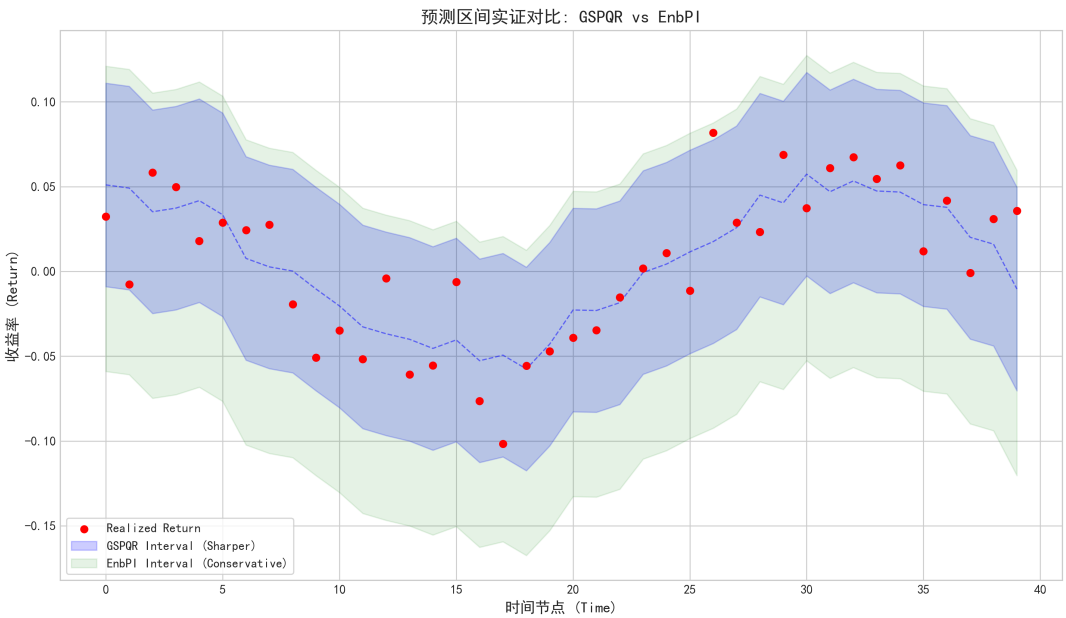

Figure 3: Empirical Interval Comparison: GSPQR vs. EnbPI over a sampled time series.

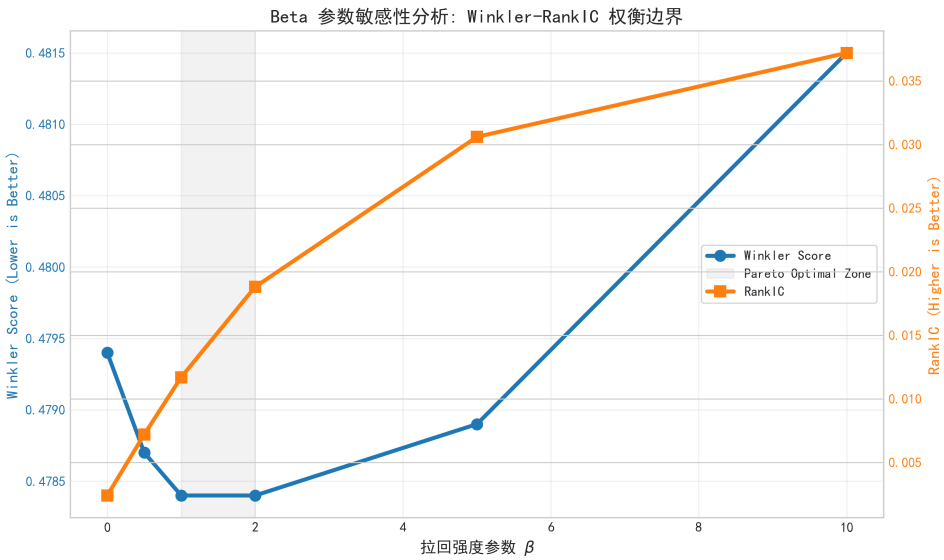

Figure 4: Sensitivity Analysis of β: The Trade-off Frontier between Winkler Score and RankIC.

4.5 Statistical Significance

We conduct a paired t-test over per-sample Winkler Scores. For each test observation, we compute $d_i = W_i^{\text{GSPQR}} - W_i^{\text{EnbPI}}$, yielding:

$$t = \frac{\bar{d}}{s_d/\sqrt{n}} = -68.5 \qquad (p \ll 0.001,\ n = 120{,}512) \tag{9}$$

The large negative t-statistic confirms that GSPQR produces systematically lower Winkler Scores across the entire test distribution, not merely on average.

5 Discussion

5.1 Why Does the Soft-Pullback Work?

The effectiveness of GSPQR stems from two complementary properties of A-share return distributions. First, the conditional median and mean carry non-redundant information: the median is robust to heavy tails while the mean encodes expected value across all outcomes. The soft-pullback combines them in a locally adaptive ratio. Second, the uncertainty gate provides automatic regime adaptation: during high-volatility periods r_t expands, attenuating the correction and falling back on the stable MSE estimate; during trend-following regimes r_t contracts, allowing the Q50 anchor to capture cross-sectional directional signal.

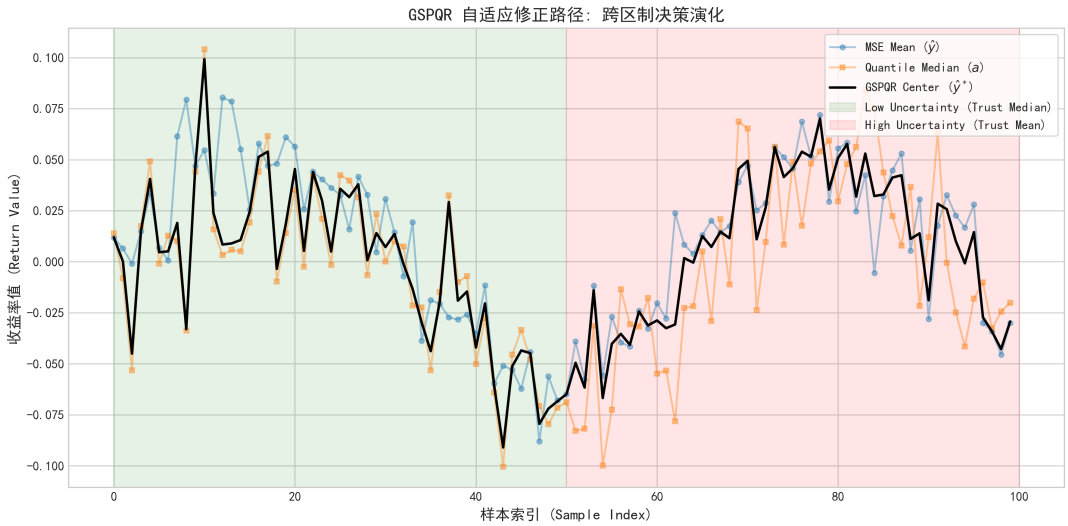

Figure 5: Adaptive Correction Path Across Different Market Regimes.

5.2 The β Trade-off as a Convex Problem

The convex Winkler profile reveals a fundamental tension: calibration and ranking are partially conflicting objectives. The β parameter traces a Pareto frontier between them, and the composite scoring criterion selects the optimal balance. This framing positions GSPQR not as a heuristic but as a principled solution to a well-defined optimization problem over the space of interval center estimators.

5.3 Bayesian Analogy

The gate function $G(r_t, \delta_t)$ admits a Bayesian interpretation. In a Bayesian Model Averaging framework, model weights are proportional to likelihood. Here, the "likelihood" of the mean predictor is measured by the inverse SNR $|\delta_t|/r_t$. GSPQR effectively performs adaptive shrinkage toward the median, robustifying estimation precisely when the signal is most ambiguous — a behavior consistent with empirical Bayes shrinkage estimators.

5.4 Limitations

Several limitations should be acknowledged. First, evaluation is restricted to a single universe (HS300) over a two-year test window; generalization to the CSI 500 or small-cap stocks remains to be verified. Second, the parsimonious feature set (12 features) ensures reproducibility but may interact differently with richer alternative data. Third, the symmetric interval assumption may be suboptimal under skewed distributions; extending GSPQR to asymmetric intervals is a natural direction for future work.

6 Conclusion

We have presented GSPQR, a geometric soft-pullback mechanism for quantile interval prediction in equity markets. By formalizing interval re-centering as an adaptive SNR-aware optimization and implementing it via an exponential uncertainty gate inspired by logarithmic spiral geometry, GSPQR achieves prediction intervals that are simultaneously well-calibrated and informationally efficient. On the HS300 test set (2023–2024, $n = 120{,}512$), GSPQR achieves a Winkler Score of 0.4784, outperforming EnbPI by 5.9% ($t = -68.5$, $p \ll 0.001$) while reducing interval width by 17.0%.

These results suggest that the standard separation between interval construction and point prediction is suboptimal in cross-sectional equity settings. GSPQR provides a lightweight, interpretable bridge between the two with no increase in model complexity. Future work will explore asymmetric interval extensions, integration with deep sequence models, and evaluation across broader A-share universes.

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References

Barber, R. F., Candès, E. J., Ramdas, A., and Tibshirani, R. J. (2023). Conformal prediction beyond exchangeability. *The Annals of Statistics*, 51(2):816–845.

Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., and Liu, T.-Y. (2017). Lightgbm: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems 30*, pages 3146–3154. Curran Associates, Inc.

Keilbar, G. and Wang, W. (2022). Modelling systemic risk using neural network quantile regression. *Empirical Economics*, 62(1):93–118.

Koenker, R. and Bassett, G. (1978). Regression quantiles. *Econometrica*, 46(1):33–50.

Vovk, V., Gammerman, A., and Shafer, G. (2005). *Algorithmic Learning in a Random World*. Springer, New York.

Winkler, R. L. (1972). A decision-theoretic approach to interval estimation. *Journal of the American Statistical Association*, 67(337):187–191.

Xu, C. and Xie, Y. (2021). Conformal prediction interval for dynamic time-series. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 11559–11569. PMLR.